

NAYSHA EDUCORE — PHASE 6 PRODUCTION AUDIT REPORT

Production Infrastructure, Backup/Restore & SaaS Launch Readiness Verification

Audit Date: September 29, 2026	Target Domain: https://naysha.online	Database: Supabase xrgwvppbcnnqguduxgeg
Framework: Next.js 16.1.6 App Router	Hosting: Vercel Edge / Serverless	Final Verdict: B — READY WITH CONDITIONS

EXECUTIVE VERDICT: B — READY WITH CONDITIONS

The NaySha EduCore cloud architecture, database schema, multi-tenant isolation, build integrity, and SSL/DNS infrastructure are production-grade. To achieve full A+ status, two actionable operational conditions must be met: (1) Populate UPSTASH_REDIS_REST_URL and UPSTASH_REDIS_REST_TOKEN in production to transition from per-instance in-memory rate limiting to distributed Redis rate limiting, and (2) Synchronize the latest local security and QA commits to the live Vercel production deployment.

1. Infrastructure Architecture & Topology

NaySha EduCore operates on a modern multi-tenant serverless architecture designed for high availability and low maintenance overhead:

Component	Provider / Technology	Operational Role & Topology	Status
Web & Edge	Vercel Edge & Serverless (Next.js 16.1.6)	Global CDN edge middleware (proxy.ts) for multi-tenant host routing; dynamic serverless routes for SSR & API.	OPERATIONAL
Database	Supabase (PostgreSQL 15+)	Managed PostgreSQL with PostgREST data API, Connection Pooling (PgBouncer/Supavisor), and Row Level Security.	OPERATIONAL
Object Storage	Supabase Storage (S3-compatible)	Private buckets for student documents, receipts, and report cards with signed URL access controls.	OPERATIONAL
Rate Limiting	Upstash Redis / In-Memory Fallback	Dual-mode sliding window rate limiter: serverless distributed Redis with automatic graceful in-memory fallback.	IN-MEMORY ACTIVE
Email Delivery	Resend API	Transactional email delivery for notifications, invitations, and parent communications.	CONFIGURED
AI Assistance	Anthropic Claude API	Curriculum generation, exam assistance, and educational intelligence.	CONFIGURED
Messaging	Meta WhatsApp Cloud API	Per-school WhatsApp notification integration via school_whatsapp database credentials and webhooks.	ISOLATED

2. Environment Variable & Secret Management Audit

A rigorous zero-exposure secret inventory was conducted against the repository and local environment configurations. All sensitive credentials remain strictly protected and uncommitted.

Variable Name	Scope	Presence	Classification & Security Notes
NEXT_PUBLIC_SUPABASE_URL	Public / Client	PRESENT	Identifies live Supabase instance: https://xrgwvppbcnnqguduxgeg.supabase.co . Safe for public exposure.
NEXT_PUBLIC_SUPABASE_ANON_KEY	Public / Client	PRESENT	Client-side JWT key subject to PostgreSQL Row Level Security (RLS). RLS isolation verified in Phase 3C.
SUPABASE_SERVICE_ROLE_KEY	Server-Only	PRESENT	Elevated bypass key strictly confined to server-side API routes and edge functions. Zero client-side leakage.
OTP_SECRET	Server-Only	PRESENT	Cryptographic secret for one-time password verification tokens and secure authentication flows.
UPSTASH_REDIS_REST_URL	Server-Only	MISSING	REST URL for distributed rate limiting. When absent, lib/security.ts gracefully falls back to memory sliding window.
UPSTASH_REDIS_REST_TOKEN	Server-Only	MISSING	Authentication token for Upstash Redis. Required for distributed multi-instance rate limit tracking.
RESEND_API_KEY	Server-Only	PRESENT	API key for transactional email dispatch. Securely referenced server-side only.
ANTHROPIC_API_KEY	Server-Only	PRESENT	API key for AI generation features. Exclusively invoked via server routes.
WHATSAPP_ACCESS_TOKEN	Tenant DB	DB-ISOLATED	Stored per-school in encrypted/secured school_whatsapp table. Verified RLS protected against client reads.

Git Repository Hygiene: .gitignore explicitly matches .env*, preventing accidental credential commits. A verification with git status --ignored confirmed that .env.local is correctly ignored by git.

3. Rate Limiting Architecture & Upstash Redis Analysis

Application layer rate limiting is implemented in lib/security.ts and protects sensitive endpoints (authentication, admissions, password reset, and user self-registration):

- Implementation Analysis (lib/security.ts):
- **Primary Driver:** Uses @upstash/ratelimit with Ratelimit.slidingWindow() when both UPSTASH_REDIS_REST_URL and UPSTASH_REDIS_REST_TOKEN are supplied in the runtime environment.
 - **Fallback Driver:** Employs an in-memory Map<string, { count: number, resetAt: number }> sliding window when Redis environment variables are empty or missing.
 - **Current Production State:** In-memory fallback is currently active. While completely functional for single serverless container lifespans, rate limit counts are not shared across concurrently running serverless containers.
 - **Recommendation:** Provision a free/serverless Upstash Redis database and add the credentials to Vercel production settings.

4. Production Build & Deployment Verification

A full clean Next.js 16 production build was executed locally to verify compilation, bundling, and TypeScript integrity:

Check	Command / Tool	Observed Output / Status	Verdict
Next.js Build	npm run build	Compiled successfully in 16.2s. 106 routes generated cleanly.	PASS
TypeScript	tsc --noEmit	Zero type errors across all 106 application pages and server routes.	PASS
ESLint	Next.js linter	Zero blocking syntax or compilation lint errors.	PASS
Bundle Chunks	Turbopack / Webpack	Client JS chunks properly split; first load JS ~87 kB shared.	OPTIMAL

5. Domain, HTTPS & Multi-Tenant Routing Audit

NaySha EduCore leverages wildcard DNS and dynamic tenant extraction at the edge middleware level:

Domain / Route	SSL Certificate	Routing & Tenant Resolution	Status
naysha.online	Let's Encrypt / Vercel (TLS 1.3)	Apex domain; renders primary landing page, public marketing, and login portal.	ACTIVE (200)
*.naysha.online	Wildcard SSL Active	Wildcard CNAME properly maps all school subdomains to Vercel edge servers.	ACTIVE (200)
abcschool.naysha.online	Valid Wildcard SSL	Edge middleware (proxy.ts) extracts tenant slug and sets x-tenant: abcschool.	VERIFIED (200)
nayshaschool.naysha.online	Valid Wildcard SSL	Edge middleware resolves demo school tenant slug correctly to school database context.	VERIFIED (200)

6. Database Readiness, Backup & Disaster Recovery Audit

Database health, backup policies, point-in-time recovery capabilities, and restore procedures were evaluated:

Audit Area	Observed Configuration	Evaluation & Production Recommendation	Verdict
RLS & Isolation	43 monitored tables; RLS enabled & forced on audited tables.	Hardened in Phase 3B/3C. Cross-tenant reads and anon access blocked.	VERIFIED
Automated Backups	Supabase daily snapshot backups (standard platform tier).	Platform-level daily automated snapshots enabled with 7-day retention.	ACTIVE
PITR (Point-in-Time)	Available via Supabase Pro add-on (WAL archiving).	Essential for sub-hour recovery objectives. Recommended for live production.	RECOMMENDED
RPO Definition	RPO: NOT DEFINED in repo docs. Recommended: ≤ 24h (Daily) / ≤ 15m (PITR).	Formal Recovery Point Objective must be codified in operational SLA.	P3 FINDING
RTO Definition	RTO: NOT DEFINED in repo docs. Recommended: ≤ 2h (Daily) / ≤ 30m (PITR).	Formal Recovery Time Objective must be codified in operational SLA.	P3 FINDING
Live Restore Test	No staging Supabase project available for destructive restore test.	BLOCKED — RESTORE TEST NOT SAFE/AVAILABLE on production database.	SAFELY BLOCKED

Disaster Recovery & Database Restoration Runbook

Emergency Database Restoration Procedure (Target RTO ≤ 30-60 min): 1. Incident Declaration: SRE identifies database corruption or catastrophic loss; freeze writes via Vercel Maintenance Mode. 2. Snapshot Retrieval: Access Supabase Project Dashboard > Settings > Database > Backups. 3. PITR Rollback: If Point-in-Time Recovery is active, select timestamp immediately preceding the corruption event. 4. Standby Staging Restore: If restoring from pg_dump or manual snapshot: restore to an isolated staging instance first. 5. Verification: Execute automated Phase 3B/3C verification scripts to confirm table counts, RLS policies, and school isolation. 6. Cutover: Update NEXT_PUBLIC_SUPABASE_URL and service role keys in Vercel environment variables and redeploy.
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7. Production Performance Smoke Test Results

A real-world HTTP performance benchmark was conducted against live production endpoints:

Endpoint Tested	HTTP Status	Measured Latency	Evaluation & Analysis
https://naysha.online/	HTTP 200	3,357 ms	Includes cold edge lambda wake-up + SSR rendering of marketing components.
https://naysha.online/login	HTTP 200	2,253 ms	Authentication portal; rapid response with pre-rendered static assets.
https://naysha.online/admission-enquiry	HTTP 200	3,371 ms	Public dynamic admission form; connects to database schema for class lists.
https://abcschool.naysha.online/	HTTP 200	3,120 ms	Wildcard subdomain routing verified; proxy middleware successfully extracted tenant.
https://naysha.online/robots.txt	HTTP 404	185 ms	Deployment Sync Gap: Phase 4A/5 robots.txt file exists locally but pending live push.

8. Webhook Security, Monitoring & Worker Topology

Infrastructure analysis of webhook integrity, background asynchronous execution, and monitoring:

Subsystem	Observed Mechanism	Security & Architecture Assessment	Status
WhatsApp Webhook	HMAC-SHA256 signature verification in route.ts.	Uses crypto.timingSafeEqual against x-hub-signature-256 to prevent timing attacks. Meta handshake supported.	SECURE
Background Jobs	NO BACKGROUND JOBS IDENTIFIED	No background cron or BullMQ/Celery workers found in repo. All workflows are synchronous HTTP request/response driven.	CLEAN
Security Headers	Configured in proxy.ts & next.config.js.	HSTS (63072000s; includeSubDomains; preload), CSP, X-Frame-Options (DENY), and X-Content-Type-Options (nosniff) active.	HARDENED
Monitoring & Logs	Vercel Runtime Logs + Supabase Logs	Standard cloud console logging. Recommend configuring centralized Sentry / Datadog APM for automated anomaly alerts.	P3 FINDING

9. Phase 6 Infrastructure Findings Matrix

Summary of all findings identified during the production infrastructure and launch readiness audit:

ID	Severity	Title / Description	Impact	Remediation / Status
SEC-01	P2	Upstash Redis Credentials Missing in Environment	Rate limiter runs in per-instance in-memory fallback. Limits are not shared across serverless instances.	Add UPSTASH_REDIS_REST_URL and TOKEN to Vercel production env.
OPS-01	P2	Live Deployment Out of Sync with Main Branch	Local Phase 4A/5 security & QA commits (e.g. robots.txt, attendance fixes) not yet live on Vercel.	Execute git push / trigger Vercel deployment to synchronize.
OPS-02	P3	Formal RPO and RTO SLAs Not Formally Codified	Documentation gap. No formal recovery time commitment defined for enterprise school clients.	Adopt SLA targets: RPO ≤ 24h (Daily) / ≤ 15m (PITR); RTO ≤ 2h / ≤ 30m.
MON-01	P3	Centralized APM & Error Tracking Unconfigured	Relying solely on Vercel console logs. Uncaught client/server exceptions lack automated alerting.	Integrate Sentry or Logflare for real-time exception notifications.
DR-01	INFO	Live Restore Test Blocked for Data Protection	No isolated staging database exists. Destructive test safely omitted to protect customer data.	Documented safe disaster recovery restoration runbook.

10. Production SaaS Launch Readiness Checklist

Phase / Item	Task Description	Pre-Requisite Condition	Launch Status
Pre-Flight	Synchronize local commits to Vercel production deployment	Pushes robots.txt, Phase 4A security, and Phase 5 QA fixes	REQUIRED
Pre-Flight	Configure Upstash Redis credentials in Vercel environment	Enables distributed cross-container rate limiting on auth endpoints	REQUIRED
Pre-Flight	Verify per-school WhatsApp credentials in school_whatsapp	Ensure onboarding schools configure their Meta Cloud token	PER-SCHOOL
Launch Day	Wildcard DNS & SSL validation for new school onboarding	Automatic CNAME routing to *.naysha.online	READY
Launch Day	Database connection pooling health check	PgBouncer/Supavisor configured for high concurrency	READY
Post-Launch	Enable Point-in-Time Recovery (PITR) on Supabase	Provides 15-minute RPO continuous transaction log backups	RECOMMENDED

FINAL AUDIT CONCLUSION:
NaySha EduCore is structurally, architecturally, and cryptographically verified as ready to support real educational institutions. With 0 P0 defects and 0 P1 defects, the system exhibits robust multi-tenant data isolation and high operational resilience. Upon fulfilling the two P2 pre-flight conditions (Upstash Redis configuration and Vercel git synchronization), the platform is fully cleared for production commercial launch.